

RNA-Seq DEG Analysis API — Overview

A production-ready web service for running Differential Expression Gene (DEG) analysis over HTTP. Built to move computational bioinformatics away from local scripts and into scalable, automated pipelines.

1. Core Architecture

The API is designed with a **Three-Tier Architecture** to handle heavy biological computations without freezing the web service.

- **API Process (FastAPI):** Handles incoming HTTP requests, authenticates users, and validates data files quickly. It never runs the analysis directly.
- **Worker Process (Procrastinate):** An asynchronous background task runner. It picks up queued jobs, executes the CPU-bound DESeq2 engine, and writes the results to the database.
- **Database (PostgreSQL):** Stores all relational data, user profiles, and manages the job queue.

2. The Data Flow

1. **Upload:** User uploads a Counts Matrix and a Metadata CSV. The API checks them immediately for errors (e.g., ensuring integer counts and matching sample names).
2. **Queue:** The user submits a job with their desired design formula. The job enters the queue.
3. **Compute:** The background worker picks up the job and runs the analysis.
4. **Results:** The worker bulk-inserts the final gene expressions into the database, making them available for the user to query, filter, or visualize (e.g., Volcano plots).

3. Designing for Predictability

- **Versioning:** All routes live under `/api/v1/` to ensure future updates don't break existing client scripts.
- **Consistent Responses:** Every API response follows a strict, predictable structure:
 - *Success:* `{ "data": { ... }, "meta": { ... } }`
 - *Error:* `{ "error": { "code": "...", "message": "..." } }`
- **Idempotency:** If a user accidentally submits the exact same job twice while the first is still running or queued, the API returns the existing job instead of starting a new one.

4. Security & Access

- **Dual Authentication:** - *Browser Users:* Short-lived JWTs with rotating refresh cookies.
 - *Scripts/Services:* SHA-256 hashed, long-lived API keys (`rnaseq_sk_...`).
- **Safety Measures:**
 - Strict upload limits (100MB max).
 - Rate limiting (e.g., 100 requests per minute).
 - Design formulas are sanitized before execution to prevent arbitrary code injection into the statistical engine.

5. Easy Deployment

The entire platform is fully containerized with **Docker Compose**. A single command orchestrates the API, the background workers, and the database, ensuring perfect parity between a local developer environment and a production cloud server.